Gargantua in the same direction as Gargantua spins (the *forward* direction). The upper red circle is a photon orbit that moves in the opposite direction to Gargantua's spin (the *backward* direction). Evidently, the whirl of space enables the forward light to be much closer to the horizon without falling in than the backward light. What a huge effect the space whirl has!

The region of space occupied by the shell of fire above and below the equatorial plane is depicted in Figure 6.4. It is a large, annular region. I omit the warping of space from this picture; it would get in the way of showing the shell of fire's full three dimensions.

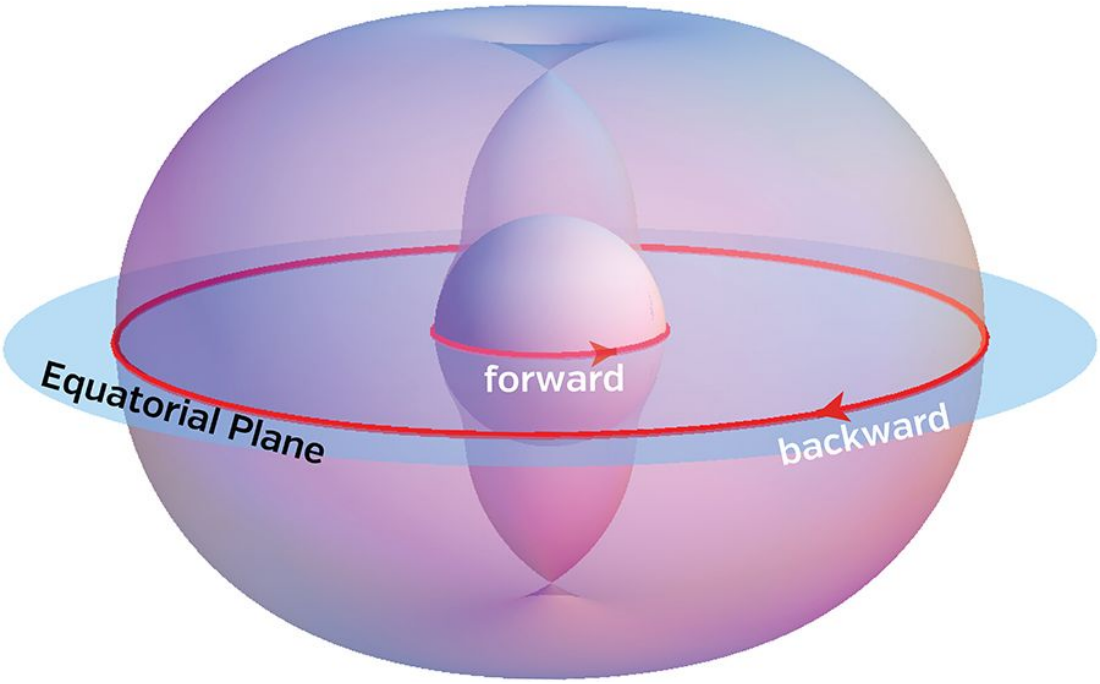



*Fig. 6.4.* The annular region around Gargantua, occupied by the shell of fire.

Figure 6.5 shows some examples of photon orbits (light rays) trapped, temporarily, in the shell of fire.

The black hole is at the center of each of these orbits. The leftmost orbit winds around and around the equatorial region of a small sphere, traveling always forward, in the same direction as Gargantua's spin. It is